

03

ENGG340 Product Design Group Project

Happy Little Pill

2022 · Teamwork (2 members)

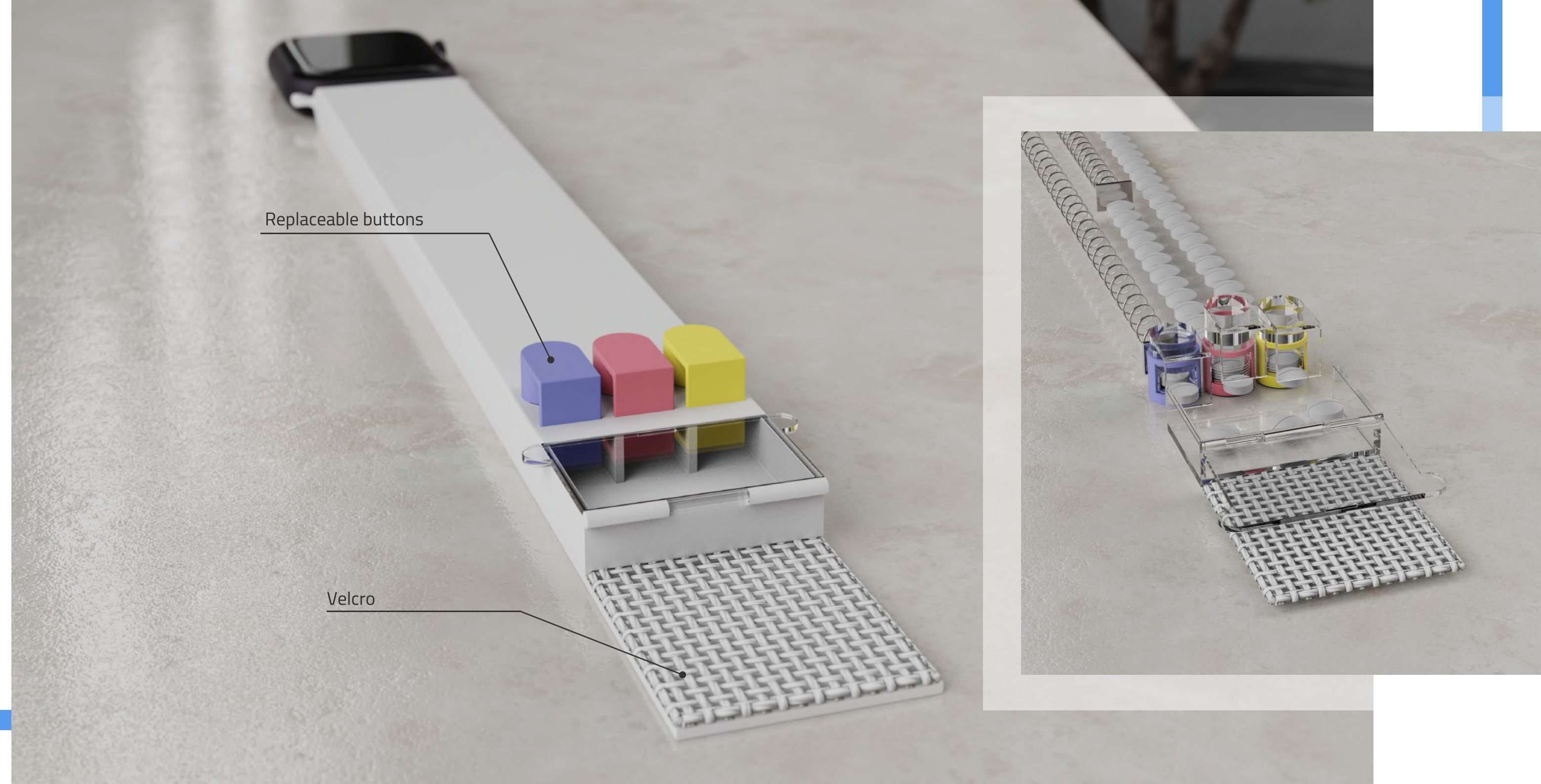
This project focuses on how to assist older people with different common illnesses to self-administer their medication.

Our group chose to investigate the issue of Alzheimer's disease on the medication of the elderly and designed an inclusive banded pill box for them to help them with their daily medication needs, while hoping that this design would raise awareness of the importance of inclusive design among designers.

Contribution: Research & Drawing & Modelling

Supervisor: Jelena Milisavljevic Syed

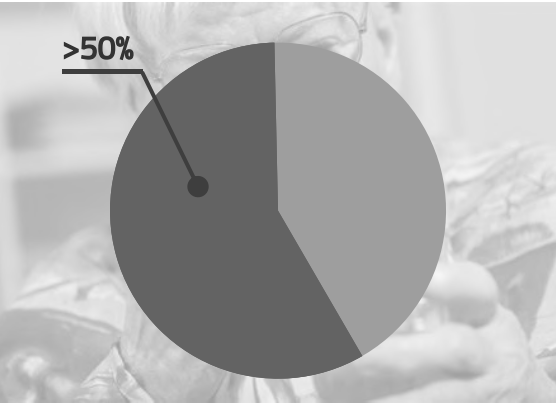
Supervisor's email: jelenams@cranfield.ac.uk



Background



Over 50% of people with Alzheimer's disease have problems taking their medication to varying degrees. As the disease progresses and worsens, the problem of taking medication becomes more pronounced.



Pills

Mistake
Forget
Reject
...



Recently the World Health Organization estimated that more than 55 million people worldwide are currently living with dementia. Approximately 50% of those with dementia are confirmed to have Alzheimer's disease.

55,000,000



Market research

Aidapt Small Weekly Pill Dispenser



Spolehli Pill Organizer



Inpher Pill Box and Cutter



Pivotell® Mk3/11 Automatic Pill Dispenser



Design Brief

Target User

People with [Alzheimer's disease](#) and other elderly people with memory impairment.

Product Define



Storage



Reminder



Emotion



Portable



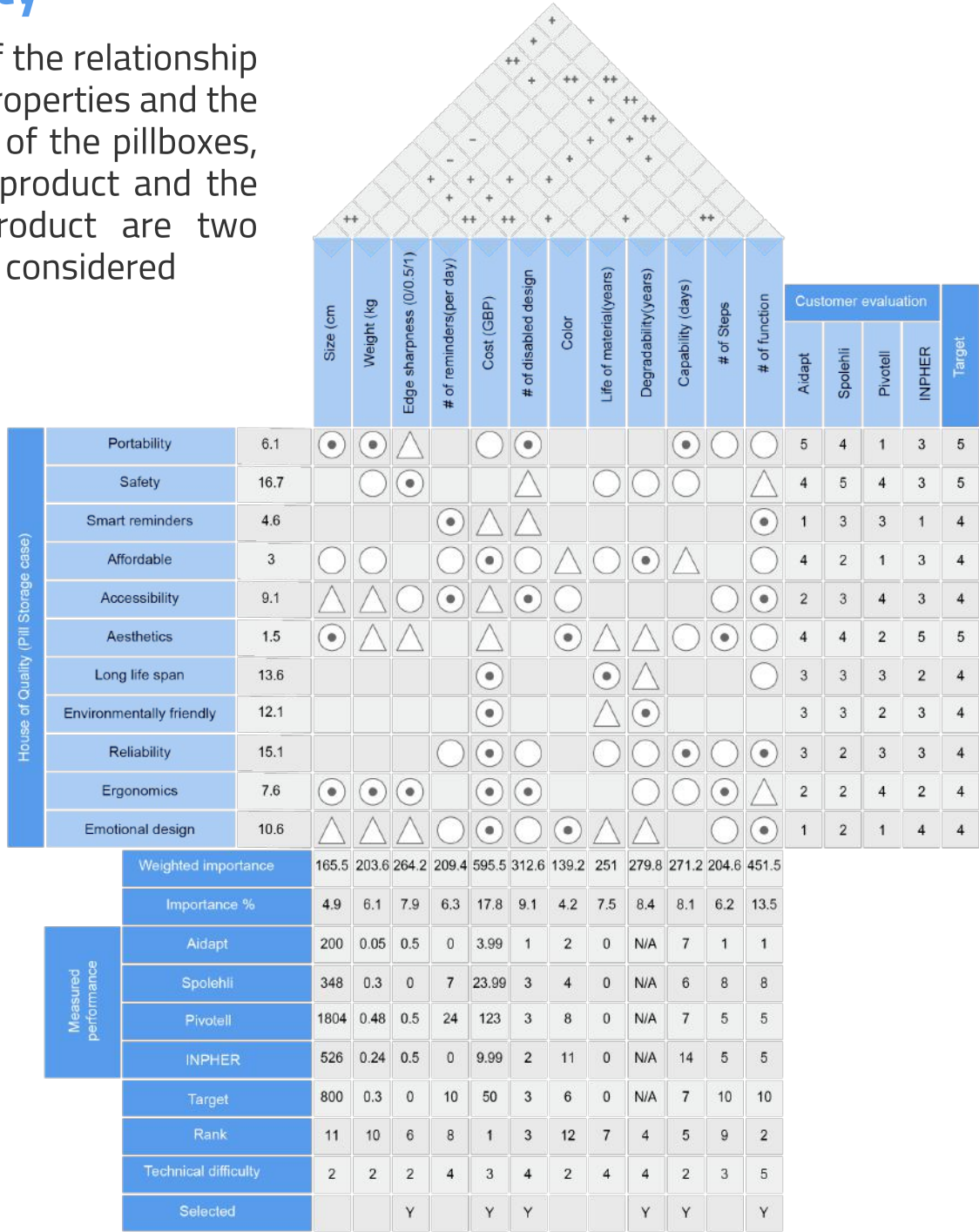
Aesthetic

House of Quality

Through the analysis of the relationship between the product properties and the industrial performance of the pillboxes, the **practicality** of the product and the **appearance** of the product are two important factors to be considered

Correlation matrix		
++	Strong positive	
+	Positive	
-	Negative	
--	Strong negative	
	Not correlated	

Relationship matrix		
●	Strong	9
○	Medium	3
△	Weak	1
	No assignment	0



Requirement list

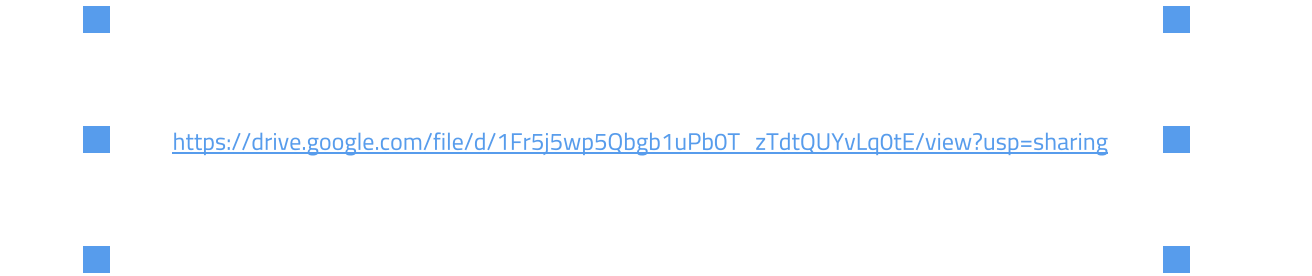
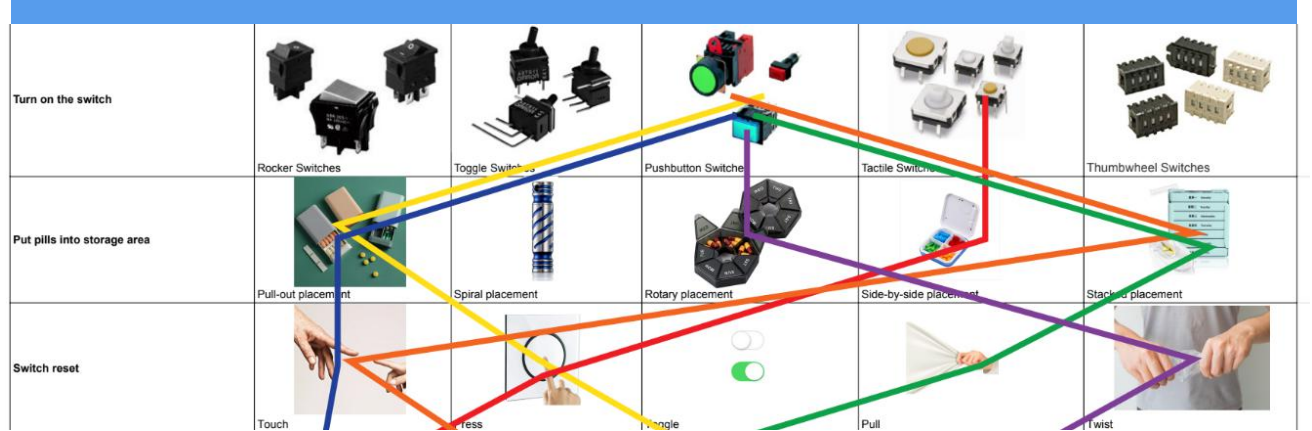
Based on the House of Quality analysis a list of requirements for this project was generated

Team: Happy little pill		Requirements List		Date:5-Nov-22	
Member A: Qunin Liu		ENGG340: Product Design Group Project			
Member B: Guanyao Chen					
Revisions	W/D	24/10/21	D	Long life span Our pillbox should have a life expectancy of no less than 10 years	A & B
Planning Requirements		24/10/21	D	Portability Size: 800cm ³ Weight: 0.3kg	A & B
30/10/21	D	24/10/21	D	Automatic reminding Our pillbox must have one or more reminder	C
30/10/21	D	24/10/21	D	Ergonomics Our pillbox must consider the appropriate ergonomics	B & C
30/10/21	D	24/10/21	D	Environmentally friendly Our pillbox must be made from recycled and green materials and must not produce environmentally or biologically harmful substances during use	A & C
30/10/21	D	24/10/21	D	Compatibility Our pillbox can use any source of power	A & B
30/10/21	D	24/10/21	D	Safety The edges of our pillbox must not be sharp	A
30/10/21	D	24/10/21	D	Affordable Our pillbox must be sold for 50 pounds or less	B
30/10/21	D	24/10/21	D	Aesthetics Our pillbox should be available in 6 colours and have a smooth shape	B
30/10/21	D	24/10/21	D	Reliability Our pill boxes should try to avoid making the elderly miss their medication	C
30/10/21	D	24/10/21	D	Increasing wiliness to take medication Our pillboxes should give the elderly emotional support and comfort	A, B & C

Preliminary

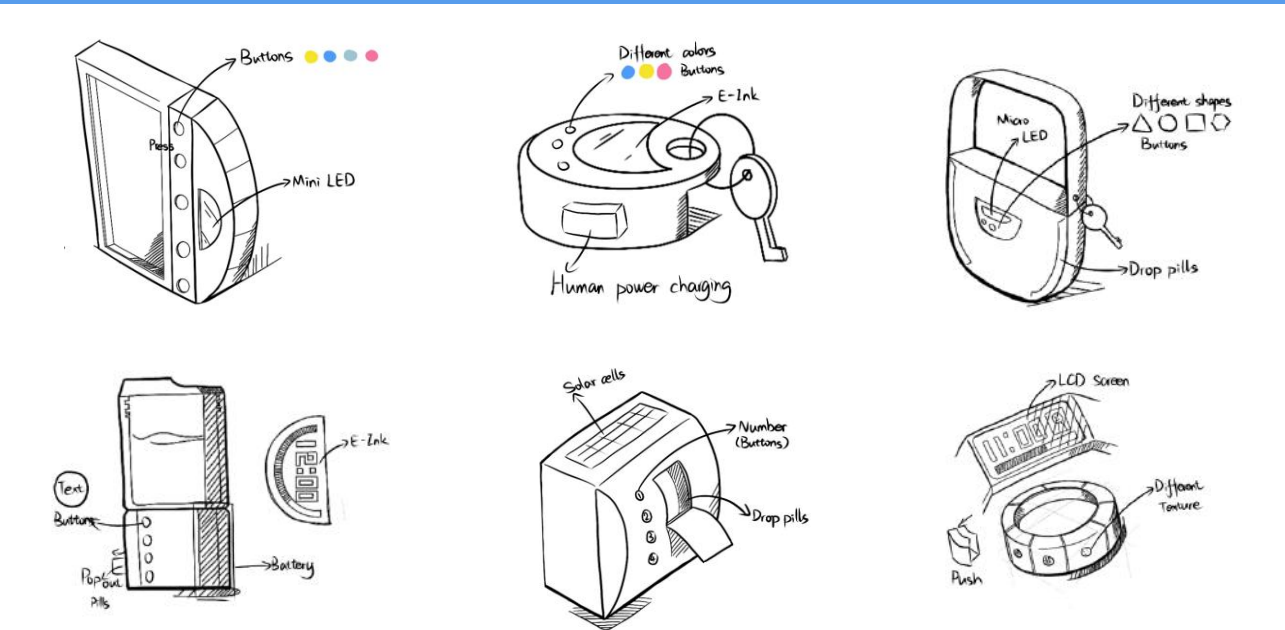
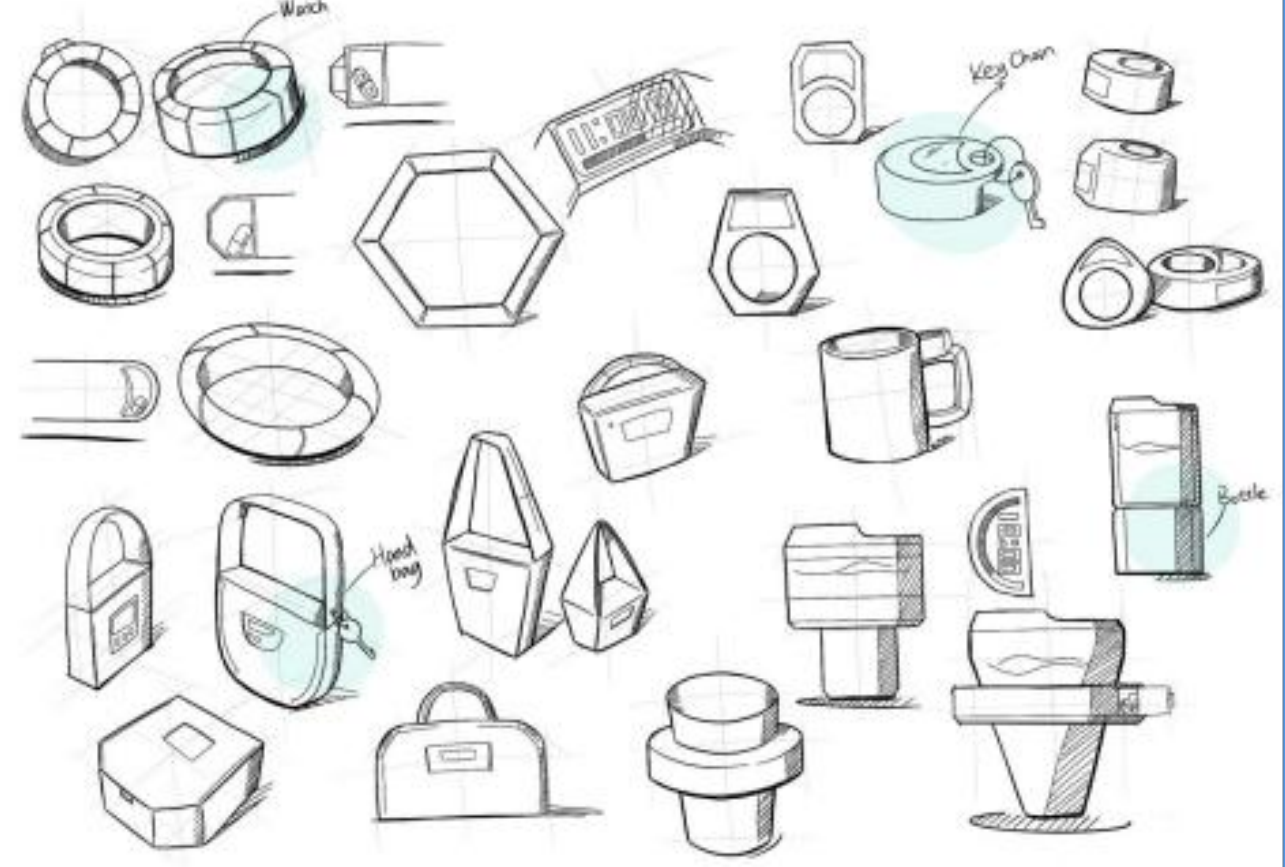
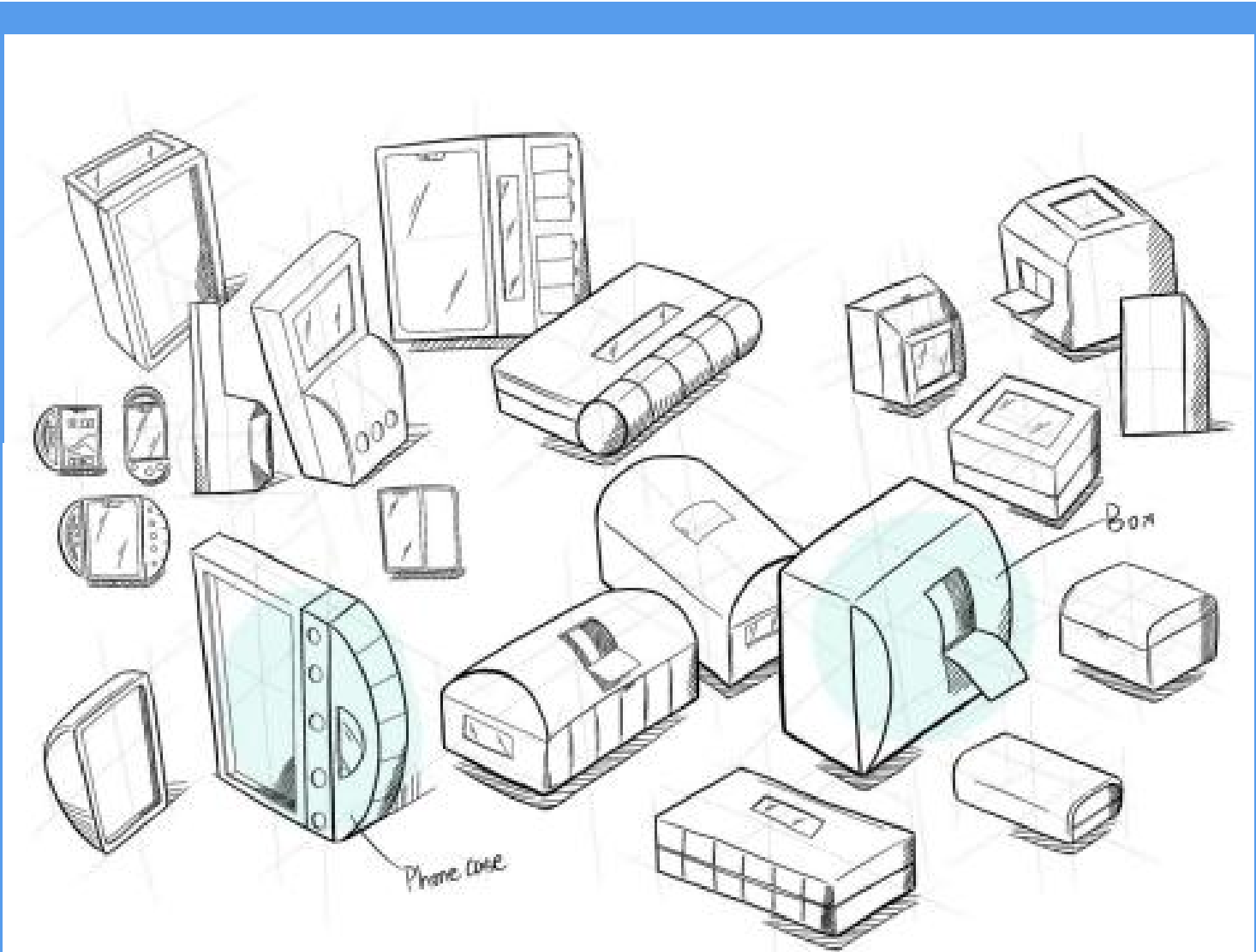
Morphological Matrix

List of 4-5 solutions based on a list of requirements and six preliminary solutions in the form of linked combinations



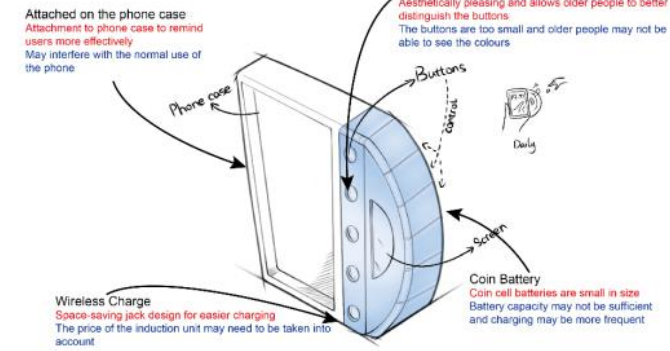
Concept Generation

List of 4-5 solutions based on a list of requirements and six preliminary solutions in the form of linked combinations

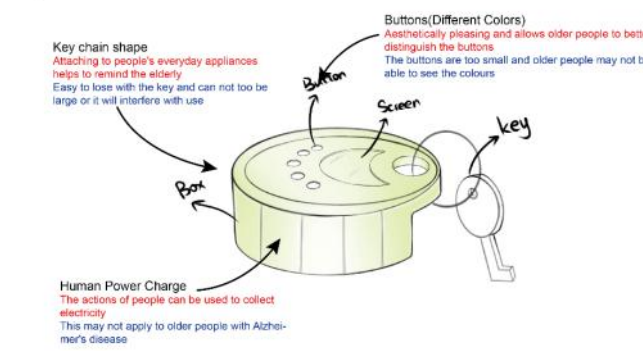


After establishing the judging rules and using the Go/NO Go Matrix for each solution, the most suitable solution is the strap pillbox

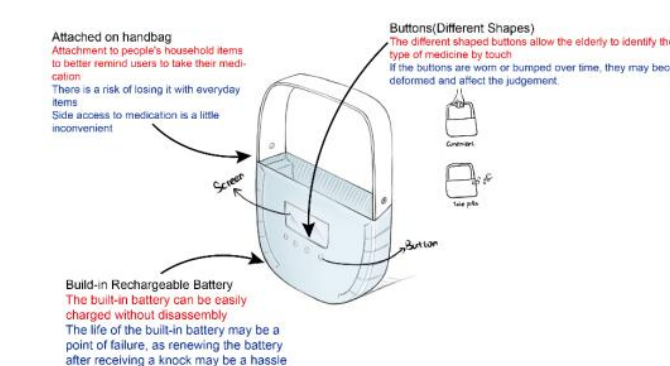
Phone case



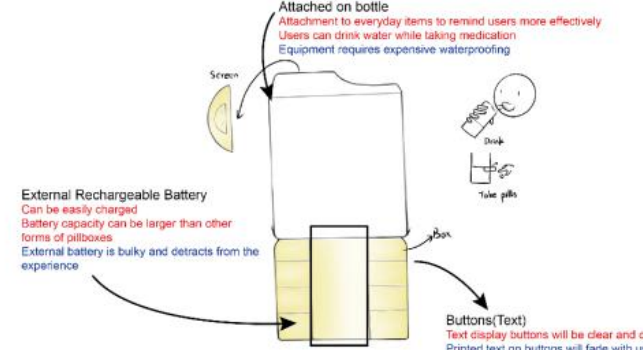
Key Chain



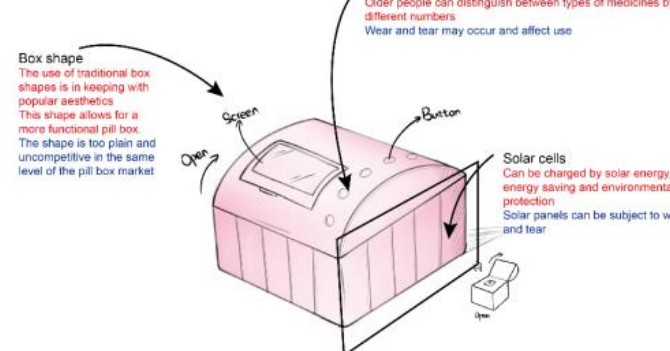
Handbag



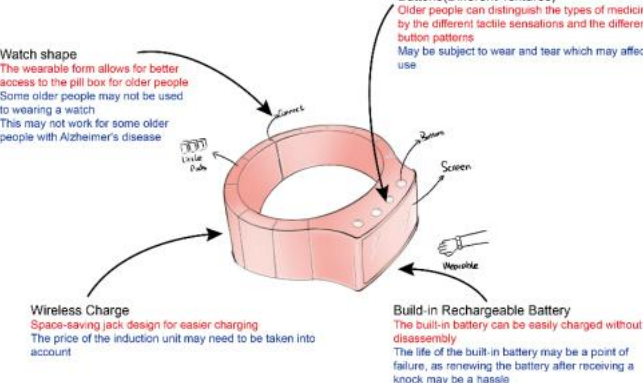
Bottle



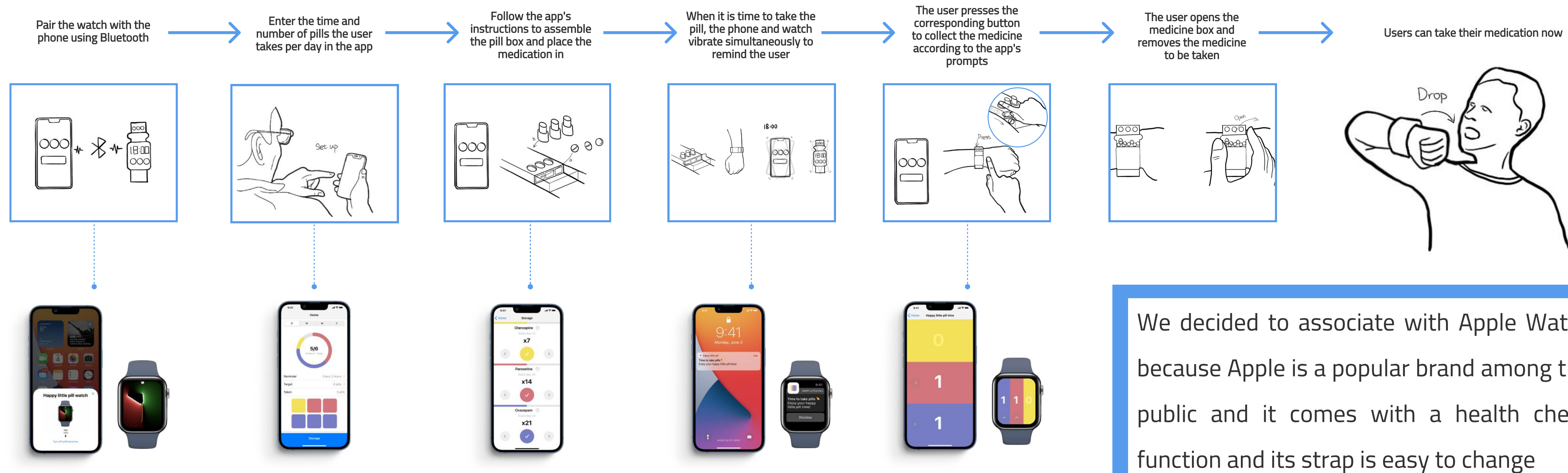
Box



Watch



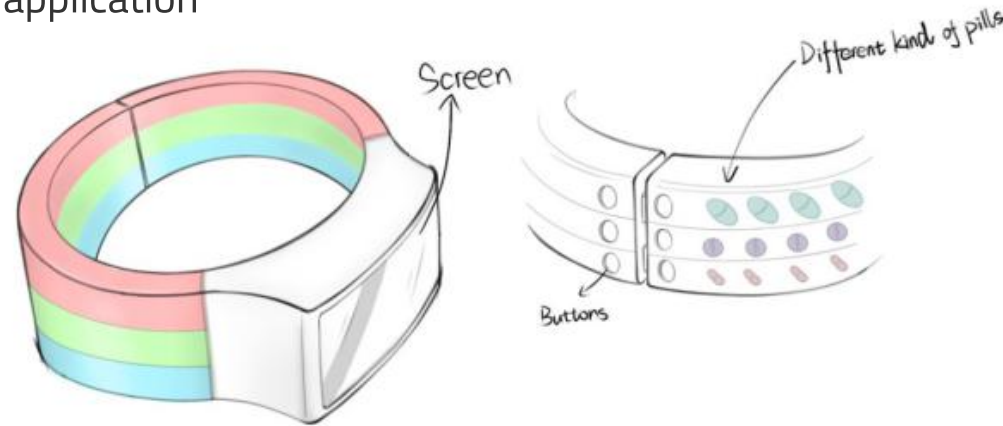
Storyboard & User Interface of the APP



Final Concept

Final Sketch

The shape of the strap pillbox is determined by its function and then the final form is determined by the practical application



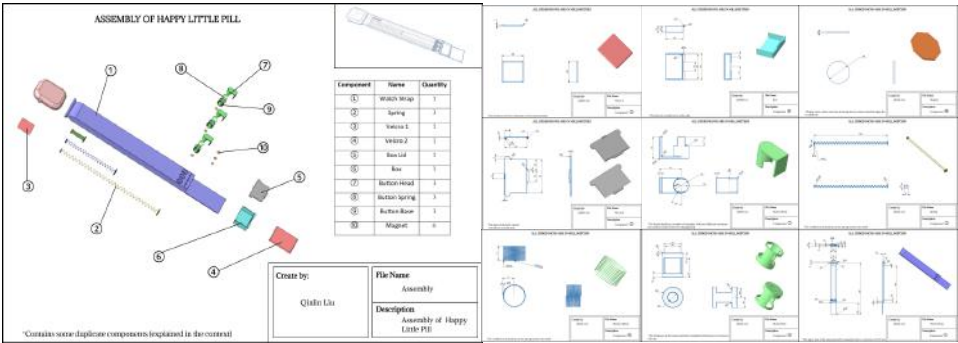
Ergonomic

The size of the pill box is determined by the size of the tablets and the average wrist length

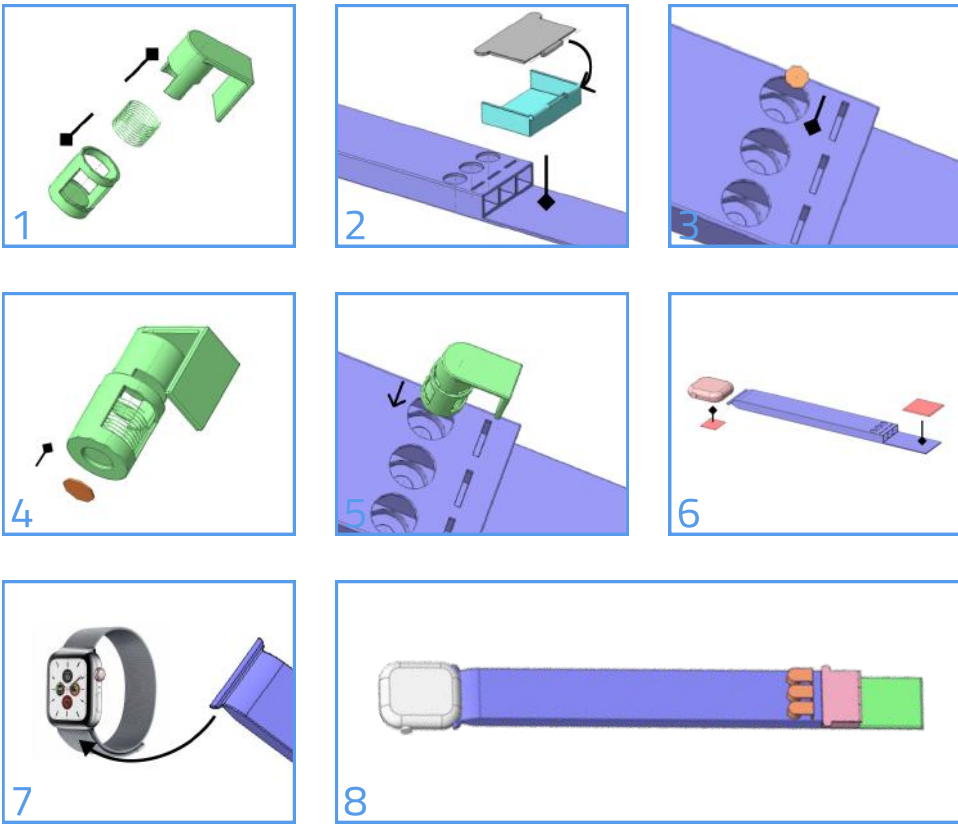


Name	One day supply	One week supply	Diameter	Required strap length
Oxazepam	3	21	0.5cm	10.5cm
Olanzapine	1	7	0.5cm	3.5cm
Paroxetine	2	14	0.5cm	7cm

Technical Drawing



Assembly



Prototype



Rendering

